Proposal ID : **OI2026B-030**
Data Received : 29/05/2026 20:00

A.M.(<=)	Any	**Moon P.**	Any/Bright
Image Quality	Any seeing os		
Cloudy Cover	Any		

Title of Proposal: ***Rotational characterization of Main Belt objects with the Zeiss telescope***

Telescope **0.60m Zeiss**

Abstract:

Studying the small bodies of the Solar System is one of the primary pathways to understanding its dynamical evolution. Objects closer to the Sun tend to be more grayish due to solar radiation erosion, thus exhibiting more uniform colors, which can also be categorized by their spectral types. Our objective is to study some of the brightest objects in the Main Belt to obtain rotation curves and define the rotational amplitude for the period of this proposal, as well as to determine the color indices for the studied objects. To this end, we request two weeks of observation time on the Zeiss telescope at the Pico dos Dias Observatory.

Category: Solar System: Small Bodies (Asteroids, Comets, Moons, Kuiper Belt, Near Earth Objects)

Time requested: 120 hours.

Minimum time accepted: 56 hours.

Observing Mode: Classical In-Person (CP)

Program Type: Director's Discretionary Time (DDT)

Principal Investigator (PI): Giuliano Margoti

PI Affiliation: ON

PI Phone/PI Email: (41)988231820 giulianomargoti@gmail.com

Co-authors

Name	Affiliation	E-mail
Matheus Margoti	UFPR	mm.matheusmargoti@gmail.com
Felipe Braga Ribas	UTFPR/DAFIS	fribas@utfpr.edu.br
Eros De Oliveira Gradovski	ON	erosgradovski@gmail.com
Julio Ignacio Bueno De Camargo	ON	camargo@on.br
Tiago Francisco Lins Leal Pinheiro	ON	franciscopinheiro@on.br
Jonatã Arcas Silva	ON	jonatasilva@on.br

Is there a thesis or dissertation that will benefit from the data?: YES

Student	E-mail	Degree	Defense
Matheus Margoti	mm.matheusmargoti@gmail.com	Undergraduate	08/2028

Optimal dates:

July 15 to July 30

Impossible dates:

nan

Instruments:

Imager (CAM 1; 2 or 4)

Detectors	- Ikon 2048 x 2048
Filters	- u Stromgren #1 (50 x 50 mm) - v Stromgren #2 (50 x 50 mm) - b Stromgren #1 (50 x 50 mm) - y Stromgren #1 (50 x 50 mm)

Scientific Justification

In the Solar System, three classes of objects are linked to in situ formation, meaning they formed in or near the orbital regions where they are currently located. These objects are the Main Belt objects, the classical Kuiper Belt objects, and the Oort Cloud objects (Nesvorny, et al. 2021).

To date, we know of only one object whose orbital adjustments place it in the Oort Cloud: comet C/2014 UN 271 (Bernardinelli–Bernstein) (Bernardinelli, et al. 2021). Its aparent magnitude in 2026 is approximately 16.5, making it nearly inaccessible to small telescopes. Known Kuiper Belt objects have absolute magnitudes between 4 and 8, with average albedos around 0.05, resulting in diameters smaller than 800 km (Fraser, et al. 2017). The magnitude of these objects is typically greater than 20, with rare exceptions, also making observation impossible in smaller telescopes.

On the other hand, objects in the Main Belt are quite numerous with varied magnitudes, reaching apparent magnitudes close to 8, such as Ceres and Vesta. Many of these objects have shapes or are oriented toward Earth in such a way that the brightness variation during the night due to their rotation is very small, which is the case for Ceres and Vesta themselves, but the set of objects in the Main Belt exceeds 1 million, providing ample options.

Therefore, our objective is to use the 60-cm Zeiss telescope at the Pico dos Dias Observatory to study bright Main Belt objects. To achieve this, we will use a wide-field camera, as we require reference stars with magnitudes close to the target body to be within the field of view.

Our intention is to characterize these objects through their rotation curves and the brightness variation as they rotate; we intend to obtain results regarding their rotation periods as well as their rotational amplitudes. Additionally, we plan to acquire images using the SDSS g, r, i, and z filters to obtain the colors of these objects and compare them with literature data and other surveys.

For this purpose, we have selected 30 candidates for observation. We have already observed a segment of the rotation curve for one of them using the 1.6-m telescope at OPD (Figure 1), and we intend to compare these results with the data obtained from the Zeiss, particularly data regarding the object's color.

Experimental Design

We intend to study bright Main Belt objects to refine their rotational properties. To this end, we have selected a set of 30 bright objects visible from the OPD with elevations greater than 15 degrees to be studied. These targets have varied rotation periods, and Table 1 shows the altitude of these objects relative to the time of night for our set of observation dates.

Our strategy is to select the targets for the night by taking into account those that will not pass too close to or over background stars, which would contaminate the object's PSF and compromise the photometry. Figure 2 shows the same object on two different days, including a day when we would not observe it due to this difficulty. We would then remain on a target for as long as possible to obtain the best possible temporal distribution. We will also observe a target on consecutive nights to complete the light curve if it was not finished the previous night. We utilize a code that predicts the rotation curve, allowing us to determine if the object will have the same face toward Earth on the following night; this helps us avoid a specific target on such a night to prevent data overlap and allow for the observation of a new target.

In addition to observing the target for the longest possible duration during the night, it is possible that the object may not be visible for a portion of the night; therefore, we have calculated that two objects would be observed per night.

The exposure time for each target will vary depending on its magnitude, but we estimate that exposure times of less than 90 seconds will be sufficient to achieve the desired signal-to-noise ratio (SNR) between 50 and 200. We plan to observe for two weeks to obtain the largest amount of data possible, aiming to analyze a significant number of objects to contribute to the final graduation work of one of the proposal's PIs.

At the beginning of the night, we will take Flat and Bias images for the calibration of the science images, moving to the targets as soon as the sun reaches -12 degrees (nautical twilight). We will set the exposure times so that our targets do not saturate or approach saturation, avoiding the non-linear flux region. Considering that we will observe with 16-bit depth, the maximum flux is slightly over 65,000; thus, we aim to maintain a peak flux near 20,000 for our target. For these observations, we will also require an autoguider to keep the field as static as possible for longer exposure times.

With the acquired images and calibrations completed, we will use the PRAIA photometry package to perform differential aperture photometry. By comparing the target with field stars, we will calculate the relative magnitude of our target, thereby obtaining the magnitude variation throughout the night.

Technical Description

We plan to use the OPD Zeiss telescope equipped with an Ikon-L camera and the SDSS g, r, i, and z filters. The Ikon-L is for providing a large field of view, allowing access to reference stars in the field that are as bright as the target. As this project is part of a graduation work, we intend to use the Zeiss telescope; with our team of observers, this will serve as a practical application of our knowledge, and since our targets are bright, they are accessible via this telescope.

Filter acquisition is also important for obtaining the color indices of the studied objects, enabling comparisons with literature and, when possible, with data from other OPD telescopes.

We request an observation period of two weeks, between July 15th and 30th. At the beginning of this period, the Moon will have nearly zero illumination, reaching full moon by the end; therefore, we will prioritize our fainter targets for the start of the campaign. We require minimum cloud cover, noting that on nights with significant cirrus, only the brightest targets can be observed. Since we aim to observe for the longest possible duration, the airmass will vary significantly throughout the night; thus, we have no restrictions in this regard.

We intend to observe for two weeks to obtain the maximum amount of high-quality data. Our exposure times will be less than 90 seconds, considering the apparent magnitude of the observed objects, in order to achieve a signal-to-noise ratio (SNR) between 50 and 200.

Figures and Tables

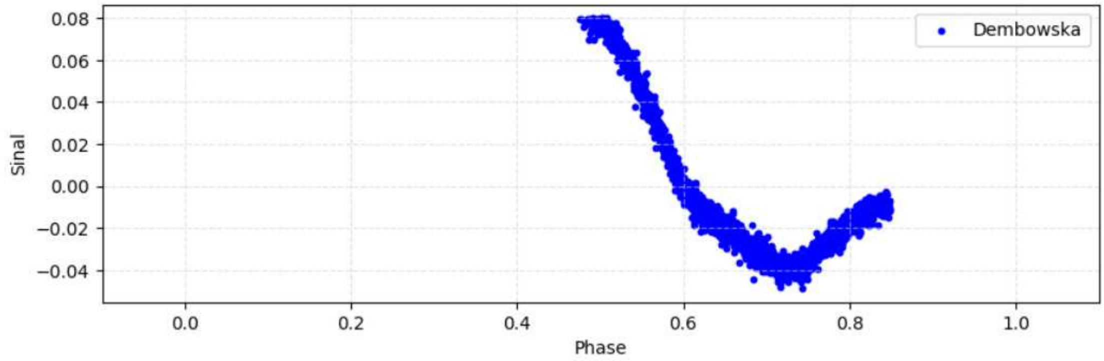

Figure 1: Segment of a light curve, equivalent to 1 hour of observation, of asteroid 349 Dembowska (A892 XB), obtained with the OPD 1.6-m telescope. In the figure, the x-axis represents the rotational phase and the y-axis represents the object signal, calculated as relative flux for the SPARC4 r' channel.

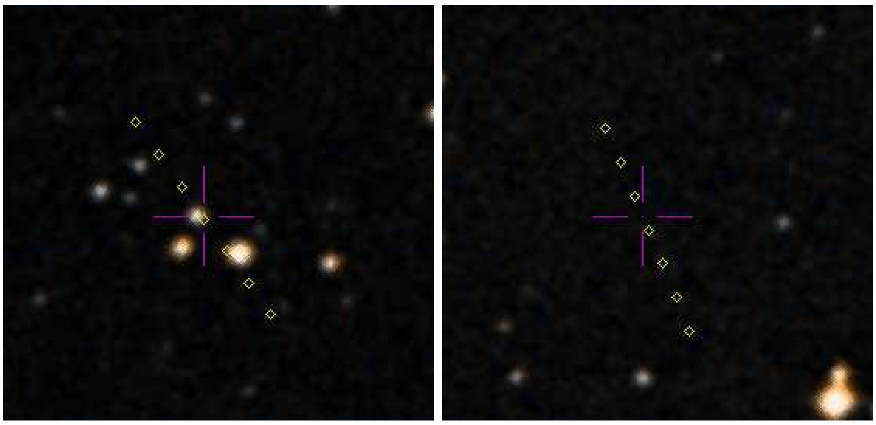

Figure 2: Left image: yellow dots represent the position of asteroid 28 Bellona (A854 EA) from the OPD with an elevation greater than 20 degrees; points are plotted every 60 minutes starting on July 16th at 18:40. Right image: similar to the left, but for July 19th at 18:40. In the left image, we can see that the target will pass very close to and over background stars, making light curve analysis unfeasible; in the right image, however, the target will pass far from background stars, which is more favorable.

Table 1: Elevation of selected Main Belt targets during the july 15 at OPD. The columns indicate the object name,the magnitude on the specific date, the rotation period in hours available in the Small Body Database (SBDB), and the hours during the night, indicating the object's elevation in degrees at each time.

Object	mag	rot	18	19	20	21	22	23	00	01	02	03	04	05	06
4 Vesta	7.7	5.3											28.6	40.3	50.2
349 Dembowska	10.1	4.7						21.4	34.0	46.9	59.9	72.8	83.5	78.1	65.6
9 Metis	10.2	5.1								29.8	42.8	55.4	66.7	73.8	70.8
354 Eleonora	10.9	4.3							29.4	42.3	54.5	65.0	70.4	67.0	57.4
21 Lutetia	11.1	8.2	34.1	47.1	60.0	72.3	80.3	74.5	62.6	49.7	36.8	23.9			
51 Nemausa	11.2	7.8			31.9	44.4	55.8	64.4	66.8	61.4	51.3	39.4	26.7		
306 Unitas	11.2	8.7						20.8	33.8	46.4	58.2	67.5	70.1	64.0	53.4
287 Nephthys	11.3	7.6						31.1	43.9	56.1	66.5	71.5	67.1	56.9	44.7
849 Ara	11.8	4.1				22.2	33.9	44.1	51.6	54.5	51.8	44.5	34.3	22.7	
13 Egeria	11.9	7.0	59.9	69.2	71.3	64.5	53.4	41.0	28.1						
389 Industria	12.1	8.5							32.1	44.5	55.9	64.4	66.7	61.2	51.1
70 Panopaea	12.1	15.8	48.4	61.3	74.1	83.8	76.6	64.0	51.0	38.1	25.4				
125 Liberatrix	12.2	4.0					28.9	41.9	54.5	65.9	73.2	70.9	61.1	49.0	36.2
28 Bellona	12.2	15.7	26.2	39.0	51.3	62.1	68.7	67.2	58.7	47.2	34.7	21.8			
480 Hansa	12.3	16.2				24.8	36.6	46.9	54.4	56.9	53.4	45.4	34.8	22.8	
135 Hertha	12.4	8.4	52.2	64.7	75.0	76.6	67.4	55.2	42.3	29.4					
258 Tyche	12.4	10.0		32.2	44.3	55.1	62.5	63.7	57.9	47.9	36.1	23.5			
24 Themis	12.5	8.4	33.2	46.2	59.1	71.6	80.8	76.0	64.1	51.3	38.3	25.4			
48 Doris	12.5	11.9	26.1	39.1	51.8	63.4	71.7	71.4	62.7	50.9	38.2	25.3			
511 Davida	12.5	5.1	42.1	38.7	31.8	22.5									
704 Interamnia	12.5	8.7	64.3	77.2	86.5	75.5	62.6	49.6	36.8	24.3					
119 Althaea	12.6	11.5	24.9	37.9	50.7	62.7	72.1	73.3	65.1	53.4	40.7	27.7			
93 Minerva	12.6	6.0	63.6	74.2	76.9	68.3	56.3	43.5	30.5						
116 Sirona	12.7	12.0	51.0	62.9	72.1	73.0	64.7	53.0	40.3	27.3					
121 Hermione	13.0	5.6	28.6	41.5	54.5	67.4	79.2	81.0	69.9	57.1	44.1	31.2			
165 Loreley	13.3	7.2										21.7	32.0	40.1	44.9
1819 Laputa	13.4	9.8					27.8	40.8	53.6	65.9	75.4	75.1	65.4	53.1	40.2
78 Diana	13.6	7.3								29.8	42.5	54.3	63.9	68.1	64.0
736 Harvard	13.7	6.7							23.5	36.4	48.9	60.1	67.9	68.2	60.7
325 Heidelberga	13.9	6.7					22.5	35.1	48.0	61.0	73.8	84.3	77.4	64.8	51.8

References

- Bernardinelli, P. H., B., et al. 2021, ApJL, 921, L37.
- Fraser, W. C., Bannister, M. T., Pike, R. E., et al. 2017, Nature Astronomy, 1, 88.
- Nesvorny, D., Li, R., Simon, J. B., et al. 2021, PSJ, 2, 27.

Previous missions of this proposal to this telescope

This proposal has not been submitted before.

Previous missions of other proposals to this telescope

In 2025B, we conducted an observation campaign at the OPD to observe the Centaur Chariklo and the Jupiter Trojan Thymbraeus using the 1.6-m telescope (between October 23rd and 27th). Since we were observing on, site as a team, we chose to arrive at the OPD earlier to observe with both the Zeiss and the 1.6-m telescopes. At the Zeiss, we observed from the 20th to the 27th, targeting two Jupiter Trojans: Mentor and Aeneas. These data are part of a broader dataset currently being prepared for submission to The Astrophysical Journal (ApJ) in 2026 for Aeneas and in 2027 for Mentor. This project was assigned to proposal P-005.

Previous results in the field by the Principal Investigator

The PI of this proposal is involved in several projects concerning the characterization of TNOs, Centaurs, and Jupiter Trojans. His Master's thesis was focused on the characterization of the TNO Quaoar and the Centaur Chariklo, titled: "Determinação da Forma Tridimensional de (50000) Quaoar por Ocultações Estelares Utilizando Algoritmo Genético para Minimização".

The papers published or submitted in recent years are:

- Size, shape, density, and atmospheric limit of (50000) Quaoar revealed from 14 years of stellar occultation
- Rigorous use of light curve inversion and stellar occultation techniques for deriving a scaled three-dimensional shape for the Jupiter Trojan (911) Agamemnon
- Evidence of a New Arc or a Belt of Small Satellites around (50000) Quaoar
- Constraining the size, shape, and albedo of the large trans-Neptunian object (28978) Ixion with multi-chord stellar occultations
- The rings of (2060) Chiron: Evidence of an evolving system
- Nsf-doe vera c. rubin observatory observations of interstellar comet 3i/atlas (c/2025 n1)
- Investigating the formation of small Solar System objects using stellar occultations by satellites: present, future and its use to update satellite orbits
- Physical characteristics of Jupiter's Trojan (1437) Diomedes from a tri-chord stellar occultation in 2020 and dimensionless three-dimensional model

PI and Co-Is publications relevant to this proposal

The co-authors of this proposal have extencive experience with rotational light curves, such as the Master's thesis presented in 2025 by Eros de Oliveira Gradovisky. His work utilized rotational and occultation data to develop irregular, concave shape models for several Jupiter Trojans: Obtenção dos Parâmetros Rotacionais e Escala de Troianos de Júpiter Através de Curvas de Rotação e Ocultações". Furthermore, our team has produced numerous studies focused on determining the physical properties of Trojans, Centaurs, and Trans-Neptunian Objects (TNOs), the main ones are:

- A ring system detected around the Centaur (10199) Chariklo
- Size and shape of Chariklo from multi-epoch stellar occultations
- Refined physical parameters for Chariklo's body and rings from stellar occultations observed between 2013 and 2020
- The size, shape, density and ring of the dwarf planet Haumea from a stellar occultation
- A Pluto-like radius and a high albedo for the dwarf planet Eris from an occultation
- Albedo and atmospheric constraints of dwarf planet Makemake from a stellar occultation

- Triplicity and physical characteristics of Asteroid (216) Kleopatra
- The size, shape, albedo, density, and atmospheric limit of transneptunian object (50000) Quaoar from multi-chord stellar occultations
- Study of the Plutino Object (208996) 2003 AZ84 from Stellar Occultations: Size, Shape, and Topographic Features
- Changing material around (2060) Chiron revealed by an occultation on December 15, 2022
- Physical characterization of double asteroid (617) Patroclus from 2007/2012 mutual events observations
- The large trans-Neptunian object 2002 TC302 from combined stellar occultation, photometry, and astrometry data
- Results from a triple chord stellar occultation and far-infrared photometry of the trans-Neptunian object (229762) 2007 UK126
- Physical properties of the trans-Neptunian object (38628) Huya from a multi-chord stellar occultation
- A stellar occultation by the transneptunian object (50000) Quaoar observed by CHEOPS
- Results of two multichord stellar occultations by dwarf planet (1) Ceres
- The Trans-Neptunian Object (84922) 2003 VS2 through Stellar Occultations
- The multichord stellar occultation on 2019 October 22 by the trans-Neptunian object (84922) 2003 VS2
- The 2017 May 20 stellar occultation by the elongated centaur (95626) 2002 GZ32
- Physical properties of Centaur (60558) 174P/Echeclus from stellar occultations

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
354 Eleonora (A893 BC)	22:00:15.42	-10:45:02.71	10.9	V	Sci	position at 2026-07-15
9 Metis (A848 HA)	23:05:42.96	-14:21:31.19	10.2	V	Sci	position at 2026-07-15
349 Dembowska (A892 XB)	22:05:47.09	-24:30:14.67	10.1	V	Sci	position at 2026-07-15
93 Minerva (A867 QA)	13:32:16.74	-17:33:48.25	12.6	V	Sci	position at 2026-07-15
849 Ara (A915 UB)	18:53:48.52	5:11:26.43	11.8	V	Sci	position at 2026-07-15
78 Diana (A863 EA)	22:53:07.18	-8:28:45.37	13.6	V	Sci	position at 2026-07-15
13 Egeria (A850 VA)	13:33:27.51	-11:43:03.75	11.9	V	Sci	position at 2026-07-15
135 Hertha (A874 DA)	14:27:16.41	-17:44:28.04	12.4	V	Sci	position at 2026-07-15
389 Industria (A894 EE)	21:39:46.25	-7:15:53.42	12.1	V	Sci	position at 2026-07-15
704 Interamnia (A910 TC)	13:46:25.35	-26:43:35.13	12.5	V	Sci	position at 2026-07-15
165 Loreley (A876 PA)	0:27:58.77	13:39:59.78	13.3	V	Sci	position at 2026-07-15
21 Lutetia (A852 VA)	15:57:12.72	-20:33:05.96	11.1	V	Sci	position at 2026-07-15
287 Nephthys (A889 QA)	20:54:49.69	-11:39:18.02	11.3	V	Sci	position at 2026-07-15
306 Unitas (A891 EA)	21:39:40.63	-10:35:17.96	11.2	V	Sci	position at 2026-07-15
480 Hansa (A901 KB)	18:47:57.44	2:48:28.92	12.3	V	Sci	position at 2026-07-15
121 Hermione (A872 JA)	16:27:30.87	-23:00:01.48	13.0	V	Sci	position at 2026-07-15
511 Davida (A903 KB)	11:31:00.71	17:41:43.60	12.5	V	Sci	position at 2026-07-15
125 Liberatrix (A872 RA)	20:09:32.15	-13:55:43.04	12.2	V	Sci	position at 2026-07-15
24 Themis (A853 GA)	16:03:00.35	-21:21:20.60	12.5	V	Sci	position at 2026-07-15
51 Nemausa (A858 BA)	17:39:56.54	-7:08:03.62	11.2	V	Sci	position at 2026-07-15
119 Althaea (A872 GA)	16:28:00.08	-14:24:41.38	12.6	V	Sci	position at 2026-07-15
28 Bellona (A854 EA)	16:11:10.16	-9:19:21.52	12.2	V	Sci	position at 2026-07-15
48 Doris (A857 SA)	16:19:27.00	-13:02:37.94	12.5	V	Sci	position at 2026-07-15
1819 Laputa (1948 PC)	20:21:24.66	-16:58:31.88	13.4	V	Sci	position at 2026-07-15
70 Panopaea (A861 JA)	14:56:40.58	-24:00:45.12	12.1	V	Sci	position at 2026-07-15
258 Tyche (A886 JA)	16:31:04.20	-4:18:36.24	12.4	V	Sci	position at 2026-07-15
325 Heidelberga (A892 EA)	21:01:33.26	-24:57:47.77	13.9	V	Sci	position at 2026-07-15
736 Harvard (A912 WC)	22:24:06.36	-9:30:54.10	13.7	V	Sci	position at 2026-07-15
116 Sirona (A871 RA)	14:25:15.40	-14:06:43.48	12.7	V	Sci	position at 2026-07-15
10999 Braga-Ribas (1978 VC6)	7:25:57.46	16:39:32.40	20.0	V	Sci	position at 2026-07-15